PRACTICAL 11

Student Management CRUD with Search and Filter

AIM

To create a student management module in StudentHub for adding, viewing, editing, deleting, searching and filtering student records using MySQLi prepared statements.

Introduction

A student management page was added to the admin dashboard. Student details are stored in the students table and displayed with pagination. The page provides validation and clear success or failure messages. Twenty sample students were added, along with CSV export and soft deletion using a status column.

Tools and Technology

VS Code, XAMPP, PHP, MySQLi, MariaDB, phpMyAdmin, HTML, CSS, browser, Git and GitHub.

Files Used

File	Purpose
students-manage.php	Student form, list, edit, delete confirmation and CSV export.
php/student-functions.php	Validation, search filters and prepared queries.
css/student-management.css	Form, table and pagination styling.
admin-dashboard.php	Link to the student management page.
sql/practical11.sql	Status column and twenty fictional seed records.
sql/practical11-students-dump.sql	Exported students table with structure and data.
scripts/test-practical11.py	Repeatable CRUD, search and access tests.
scripts/export-practical11.php	Generates the students-only SQL dump.

Key Skills Used

PHP-MySQL CRUD, prepared statements, validation, search, filters, pagination, admin access, CSV export, soft deletion and user feedback messages.

Implementation Overview

1 Database and Sample Records

The existing students table stores student ID, full name, email, mobile, course, year and creation time. A status column was added with active and deleted values. The seed file adds twenty fictional students while retaining the existing records.

2 Add and View Students

The Add student form validates the entered details and uses a prepared INSERT statement. A SELECT query loads the student list. Duplicate emails and invalid fields are rejected with appropriate messages. Only admin accounts can access this module.

3 Edit Student Details

Edit opens the selected record in the form. Saving the changes uses a prepared UPDATE statement. The student ID and email are checked, and a success message confirms the update.

4 Delete Confirmation and Soft Delete

Delete opens a confirmation page displaying the student name and email. The user must confirm before submitting. The advanced soft-delete option uses UPDATE to set status to deleted instead of physically removing the row. The record is hidden from the active list and can be restored.

5 Search and Filters

Search works on name, email and course, including displayed course names such as B.Tech IT. Course, year and status filters can be combined. Prepared parameters are used for search values, and no-results searches display a clear message.

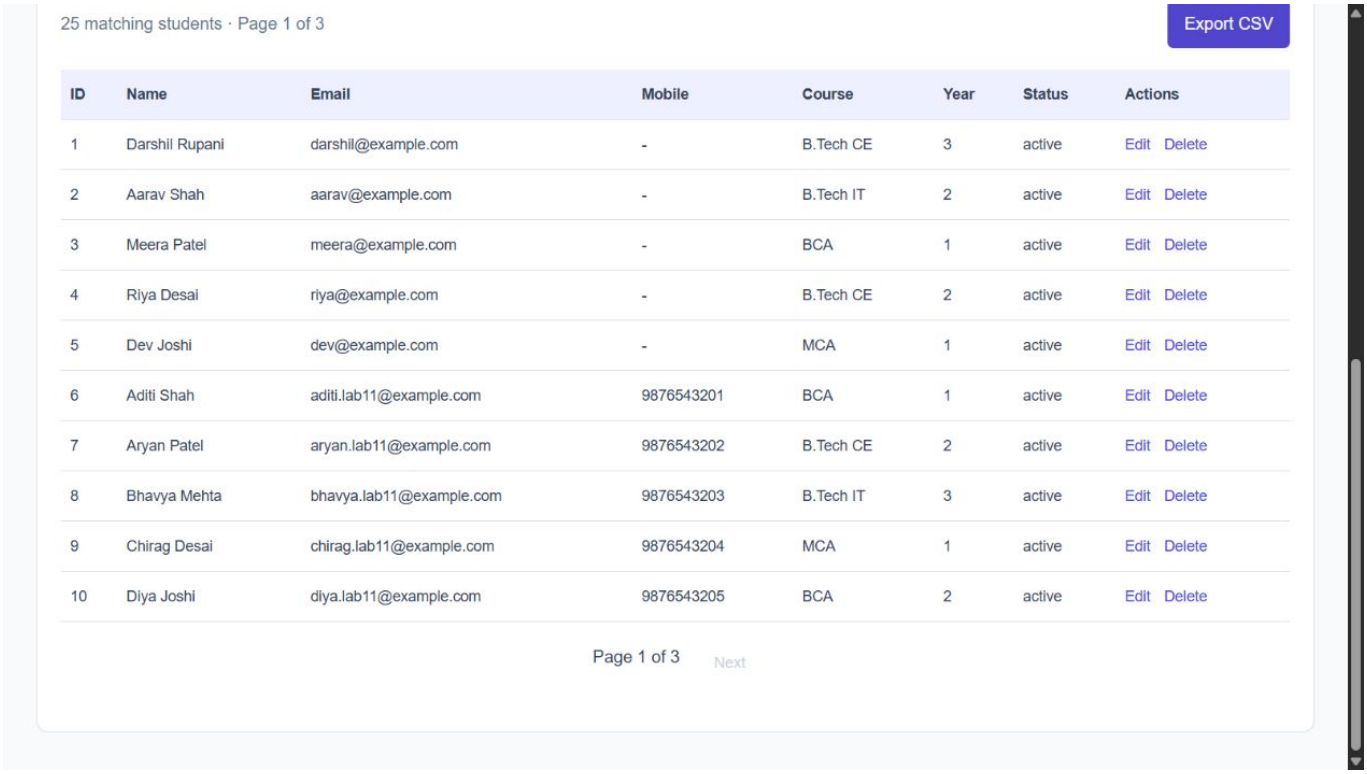
6 Pagination and CSV Export

The list displays ten records per page. Previous and Next links preserve the current search and filters. Export CSV downloads all matching records across pages. Values that could be interpreted as spreadsheet formulas are escaped.

7 SQL Dump and Testing

The student table structure and records were exported to a separate SQL file. The dump was imported successfully into a temporary table for verification. CRUD operations, invalid inputs, duplicate emails, access checks, search, filters, pagination and CSV export were tested.

Screenshots of Student List and Add Operation



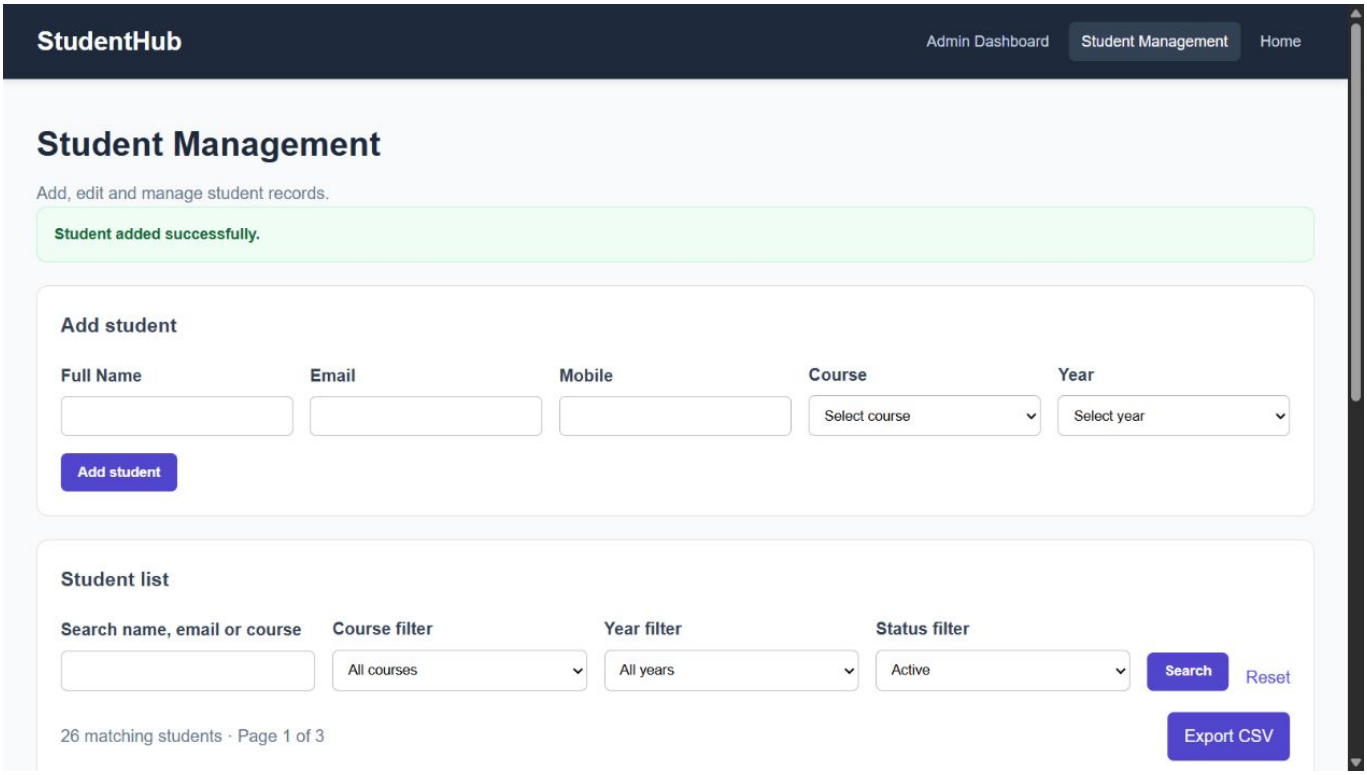
25 matching students · Page 1 of 3

Export CSV

ID	Name	Email	Mobile	Course	Year	Status	Actions
1	Darshil Rupani	darshil@example.com	-	B.Tech CE	3	active	Edit Delete
2	Aarav Shah	aarav@example.com	-	B.Tech IT	2	active	Edit Delete
3	Meera Patel	meera@example.com	-	BCA	1	active	Edit Delete
4	Riya Desai	riya@example.com	-	B.Tech CE	2	active	Edit Delete
5	Dev Joshi	dev@example.com	-	MCA	1	active	Edit Delete
6	Aditi Shah	aditi.lab11@example.com	9876543201	BCA	1	active	Edit Delete
7	Aryan Patel	aryan.lab11@example.com	9876543202	B.Tech CE	2	active	Edit Delete
8	Bhavya Mehta	bhavya.lab11@example.com	9876543203	B.Tech IT	3	active	Edit Delete
9	Chirag Desai	chirag.lab11@example.com	9876543204	MCA	1	active	Edit Delete
10	Diya Joshi	diya.lab11@example.com	9876543205	BCA	2	active	Edit Delete

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Figure 1: Database student list displaying the first page of records.



StudentHub

Admin Dashboard | Student Management | Home

Student Management

Add, edit and manage student records.

Student added successfully.

Add student

Full Name | Email | Mobile | Course | Year

Select course | Select year

Add student

Student list

Search name, email or course | Course filter | Year filter | Status filter

All courses | All years | Active

Search Reset

26 matching students · Page 1 of 3

Export CSV

Figure 2: Valid student details added successfully.

Screenshots of Edit, Search and Filter

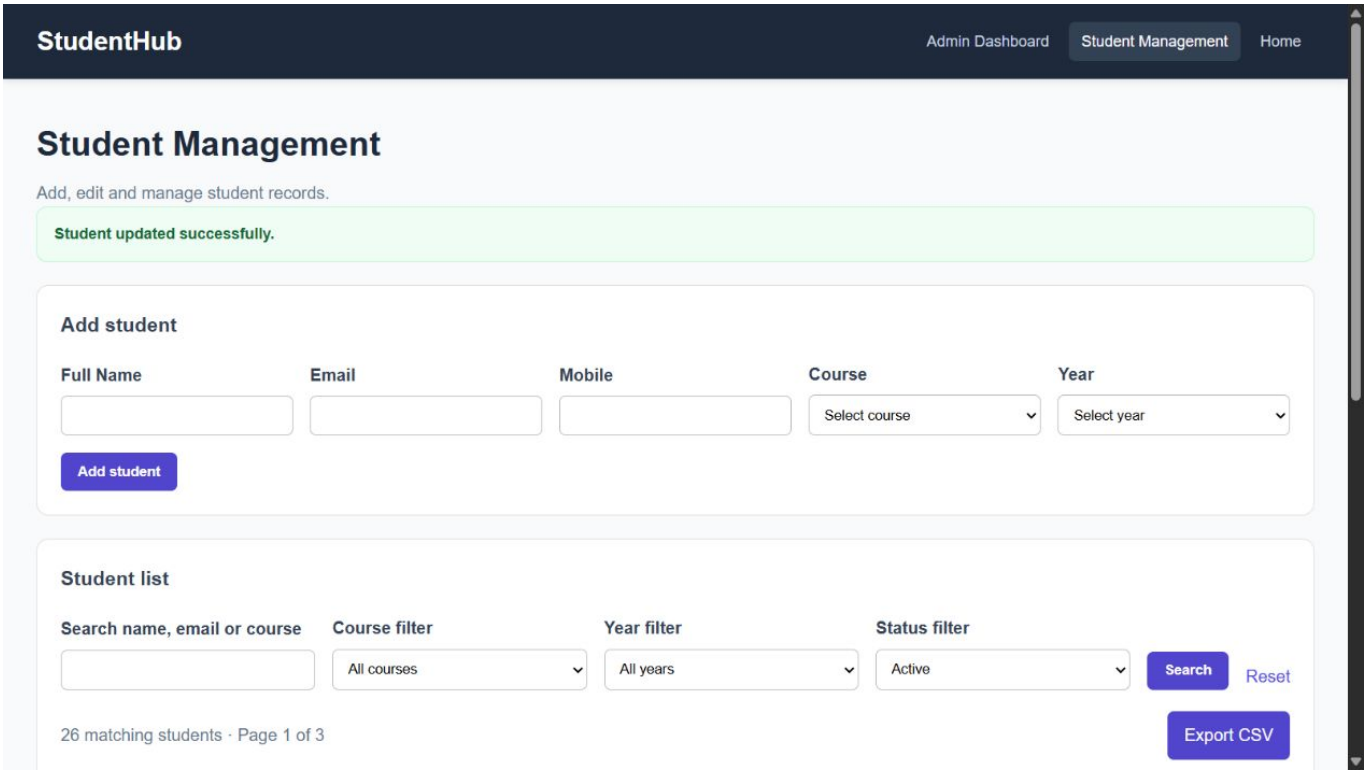


Figure 3: Student details updated successfully.

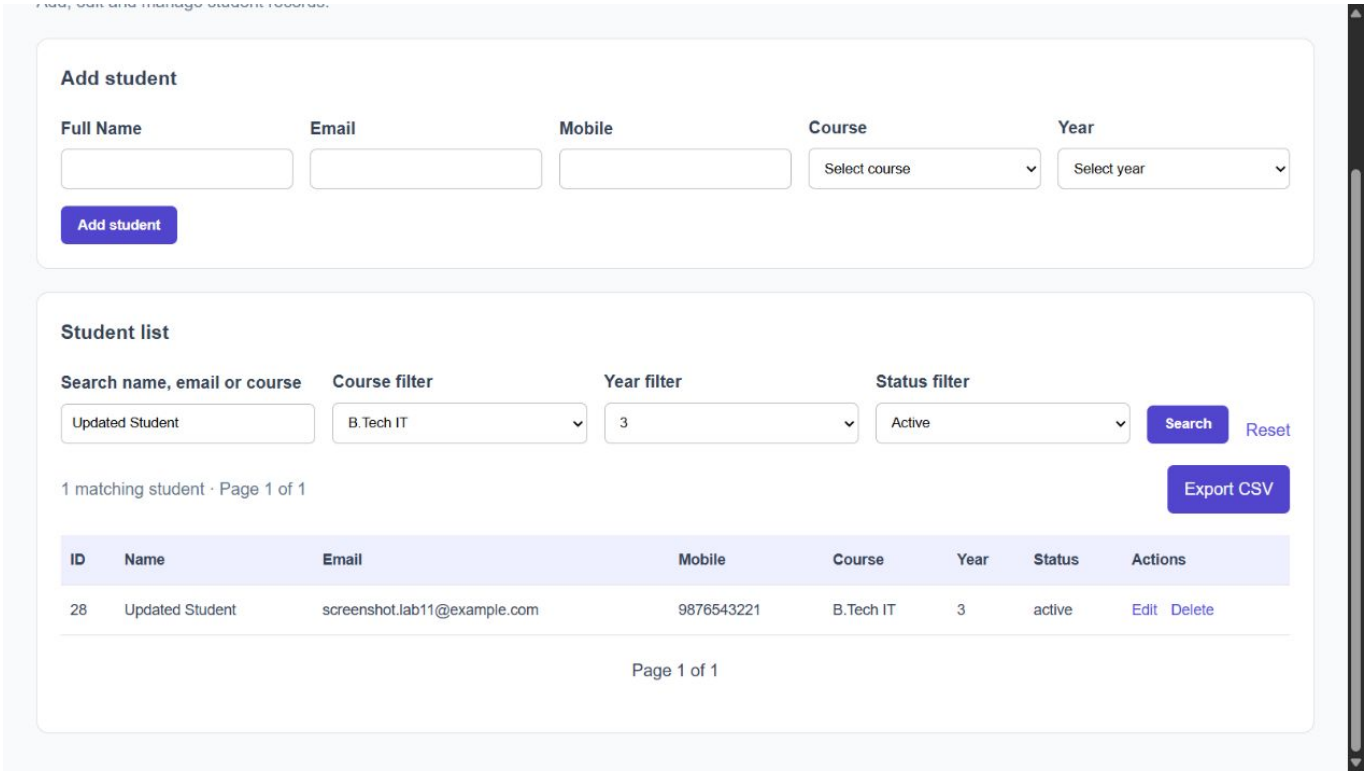


Figure 4: Search by name combined with course and year filters.

Screenshots of Delete Confirmation and Result

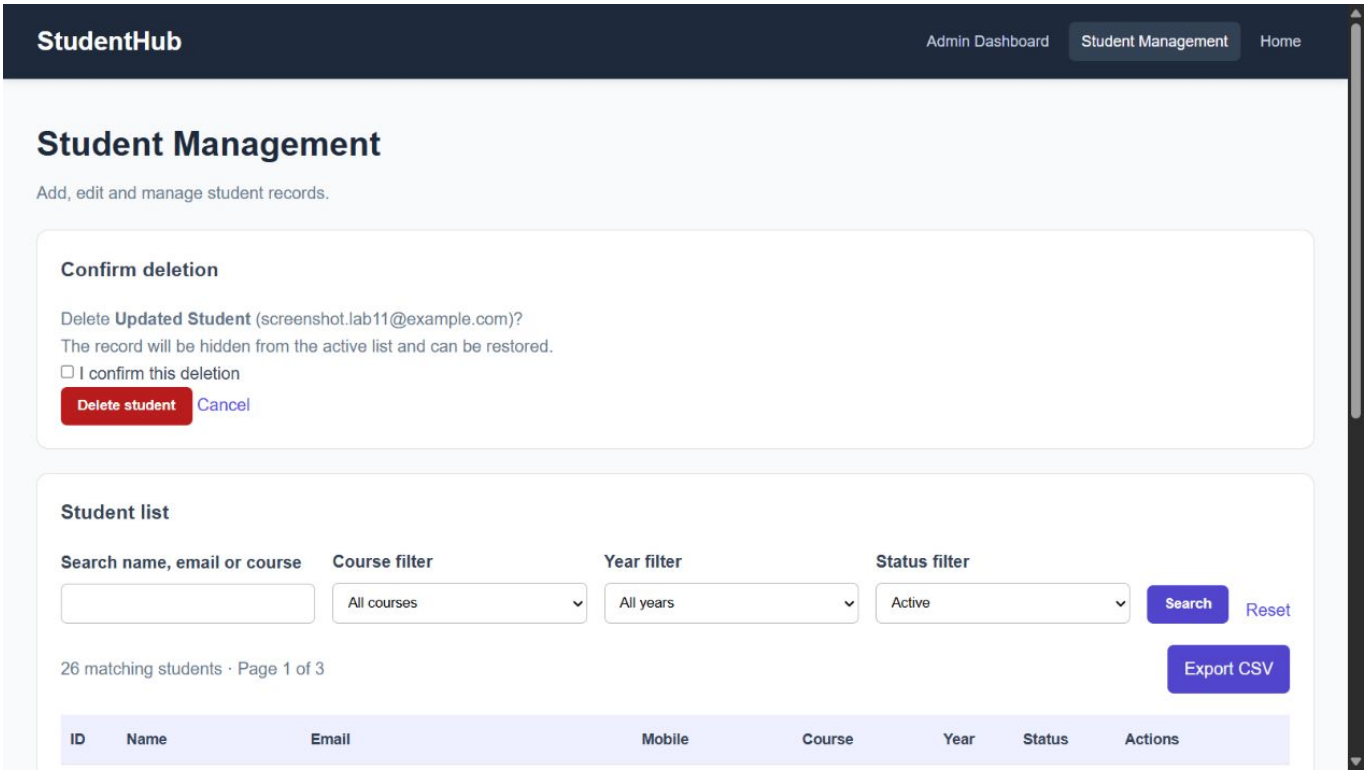


Figure 5: Confirmation is required before deleting the selected student.

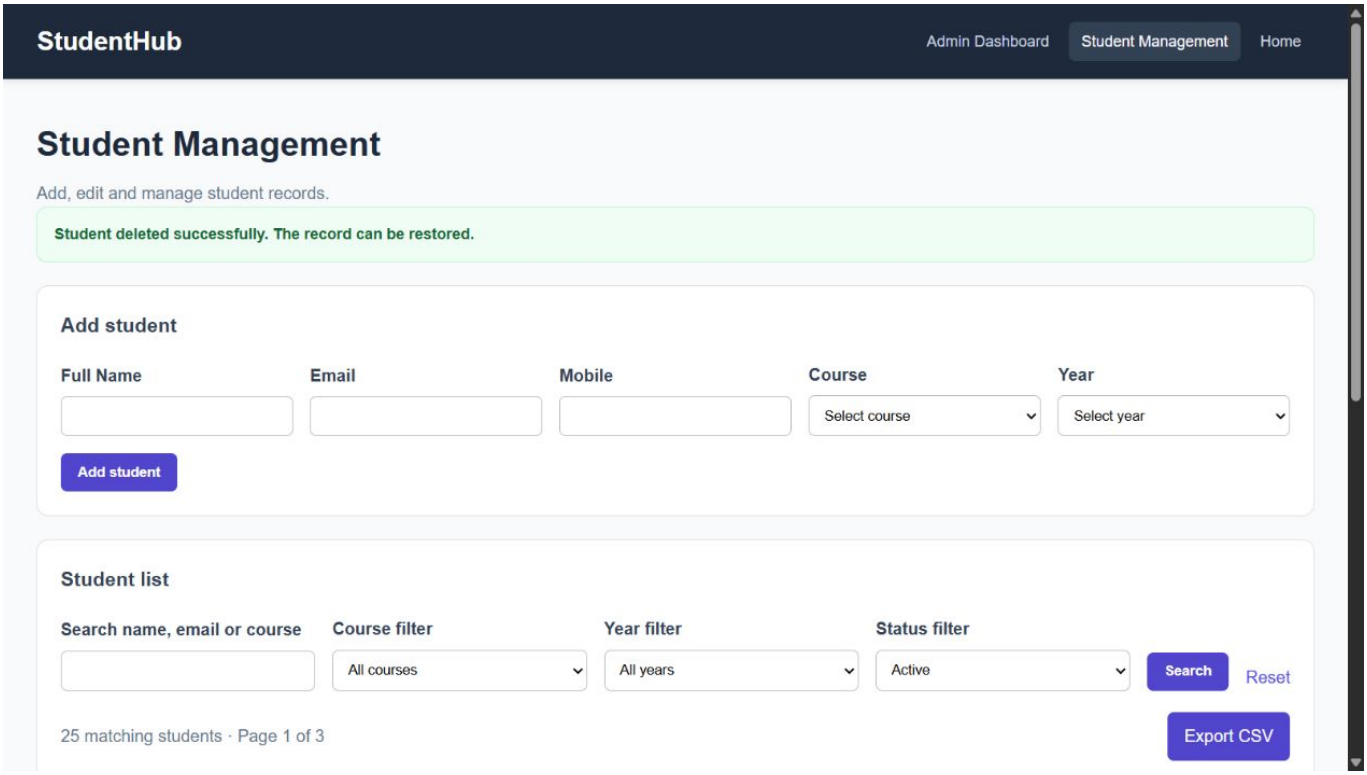


Figure 6: Successful soft deletion with a clear confirmation message.

Screenshots of Soft Delete and Pagination

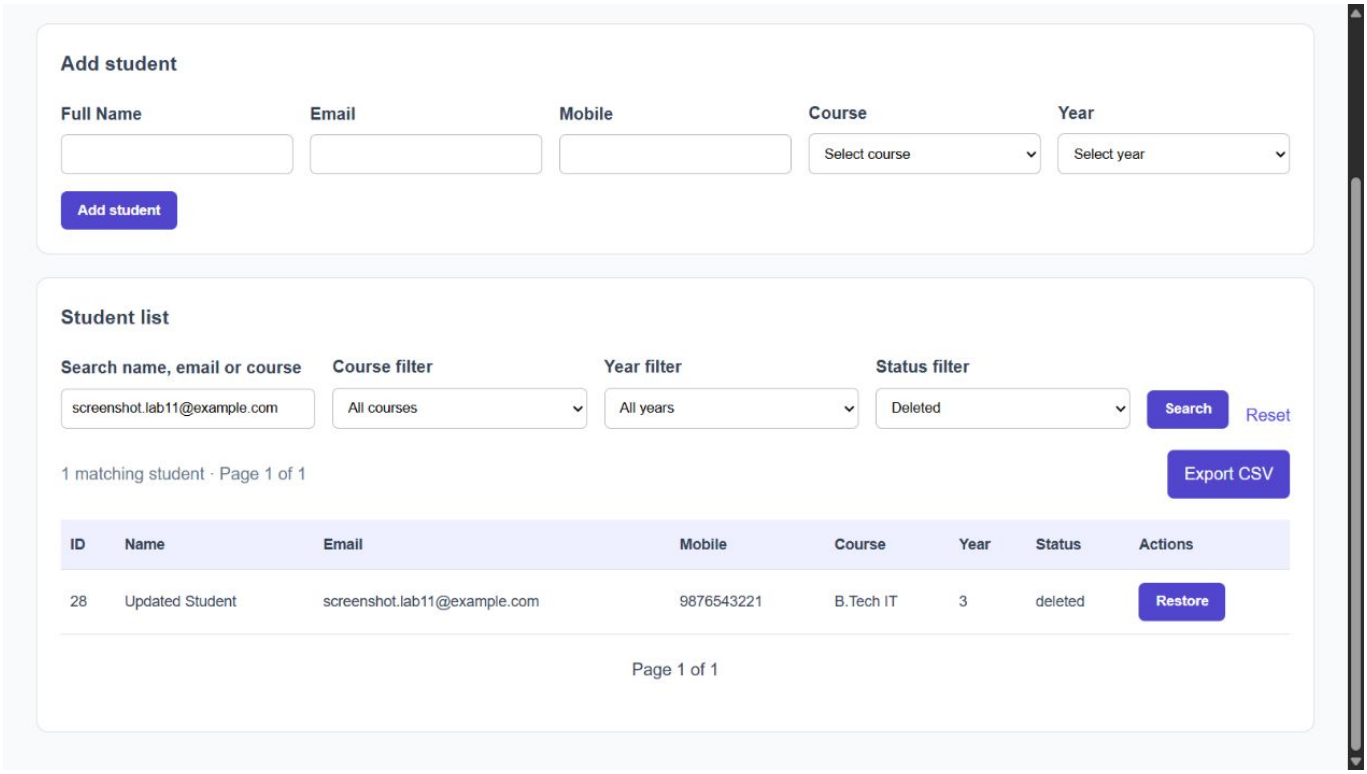


Figure 7: The deleted record remains stored and has a Restore option.

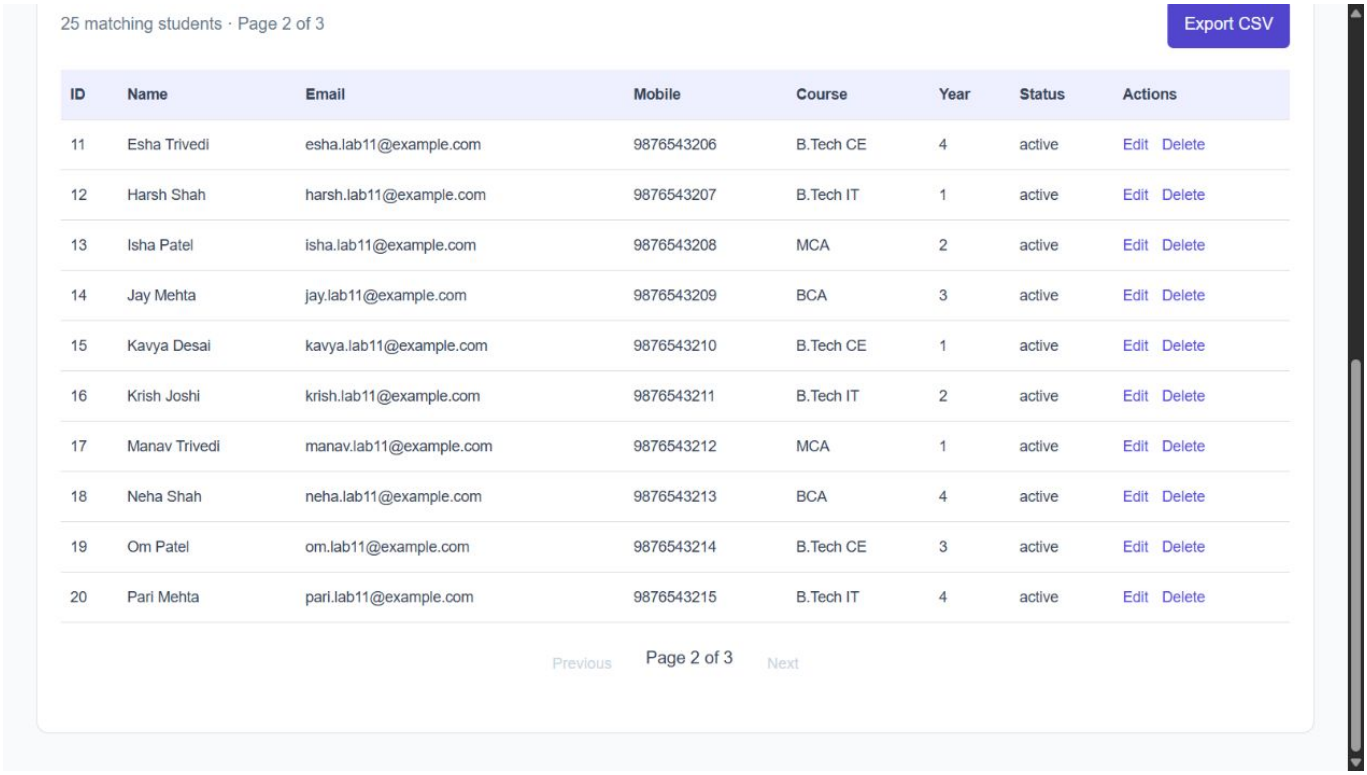


Figure 8: Second page of student records with Previous and Next links.

Database and Export Results

Check	Observed result
New fictional seed records	20
Active students after testing	25
Records in SQL dump	29
SQL dump verification	Imported successfully into a temporary table.
Sample CSV	25 active students; seven columns.

Testing Summary

Test input or action	Observed result
Admin / student / anonymous access	Allowed / HTTP 403 / login redirect.
Add valid student	Student added successfully.
Duplicate email / invalid fields	Rejected with clear validation messages.
Edit student	Updated name, course and year saved.
Search by name / email / course	Matching records returned.
Displayed course name search	B.Tech IT matches its stored course code.
Course and year filters	Only matching course/year records returned.
Pagination / empty search	Correct page / No students found.
SQL text / literal percent search	Treated as search data; no unexpected rows.
Delete without confirmation	Rejected; record retained.
Confirmed deletion	Status changed to deleted.
Deleted status / Restore	Record remains available and can be restored.
CSV export	All matching records exported across pages.
Invalid CSRF / missing student	HTTP 403 / clear not-found message.

All 26 HTTP checks passed. The twenty seed records and the existing sample students remain active. The four demonstration records remain soft-deleted.

Key Questions and Answers

Q1 Are SELECT, INSERT, UPDATE and DELETE operations working

SELECT loads the list, INSERT adds students and UPDATE saves edits. The delete action works through the advanced soft-delete approach: status is updated to deleted instead of issuing a permanent SQL DELETE. Add, edit, delete and restore tests passed.

Q2 Is search by name/email/course implemented

Yes. A prepared query searches name, email and course. The displayed course name is also supported. Course, year and status filters can be applied together, and pagination retains the selected filters.

Q3 Are delete operations confirmed before execution

Yes. Clicking Delete opens a confirmation page with the selected student details. A confirmation checkbox and CSRF token are required before the POST request changes the status. Submitting without confirmation was rejected.

GitHub Repository

https://github.com/DarshilRupani/studenthub-portal
Practical 11 source commit: d16c903
Message: Complete Practical 11 student CRUD search filters and soft delete

Post Laboratory Work

Submit the student management source files, sql/practical11-students-dump.sql, the CRUD screenshots and test cases in this report. The migration and seed file is sql/practical11.sql. A sample CSV is saved in output/csv/student-list-practical11.csv.

Conclusion

The student management module was completed using MySQLi prepared statements. Add, view, edit, confirmed deletion, search, filters and pagination worked correctly in the tests. The intermediate CSV export and advanced soft-delete extensions were completed. The SQL dump was exported and verified successfully.